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ABSTRACT

The Hospital Management System is a database-driven application designed to efficiently manage hospital operations such as patient records, doctor details, appointments, treatments, and billing.This project uses SQL concepts including table design, constraints, joins, aggregate functions, transactions, stored procedures, triggers, and views to ensure data consistency and automation. A graphical user interface (GUI) is integrated with the database, allowing users to interact with the system for real-time data management. The system improves efficiency, reduces manual errors, and provides analytical insights through reports.

CHAPTER 1: INTRODUCTION

Hospitals require efficient systems to manage large volumes of patient and operational data. Manual methods are prone to errors, delays, and inefficiencies.
This project implements a **Hospital Management System using SQL and GUI**, which automates key operations such as:

- Patient registration
- Appointment scheduling
- Treatment tracking
- Billing management
- Reporting and analytics

The system ensures data accuracy, consistency, and ease of access.

CHAPTER 2: PROBLEM STATEMENT & OBJECTIVES

◆ Problem Statement

Manual hospital management systems are inefficient, error-prone, and difficult to scale. There is a need for a centralized system to manage hospital data effectively.

◆ Objectives

- Design a relational database for hospital operations
- Implement CRUD operations using SQL
- Use advanced SQL features (procedures, triggers, views)
- Ensure data consistency using transactions
- Integrate database with GUI for user interaction
- Generate reports for decision making

CHAPTER 3: METHODOLOGY

The project follows a structured approach:

1. **Database Design**
   - Identified entities: Patient, Doctor, Appointment, Treatment, Billing, Department
2. **Schema Creation**
   - Created tables with constraints (Primary Key, Foreign Key, CHECK)
3. **Data Insertion**
   - Inserted sample data to simulate real-world hospital operations
4. **Query Implementation**
   - Developed queries for data retrieval and reporting
5. **Advanced Features**
   - Implemented transactions, cursors, triggers, and stored procedures
6. **GUI Integration**
   - Connected SQL database with GUI for user interaction

CHAPTER 4: ER DIAGRAM & RELATIONAL TABLES

🔷 Entities:

- Patient
- Doctor
- Department
- Appointment
- Treatment
- Billing

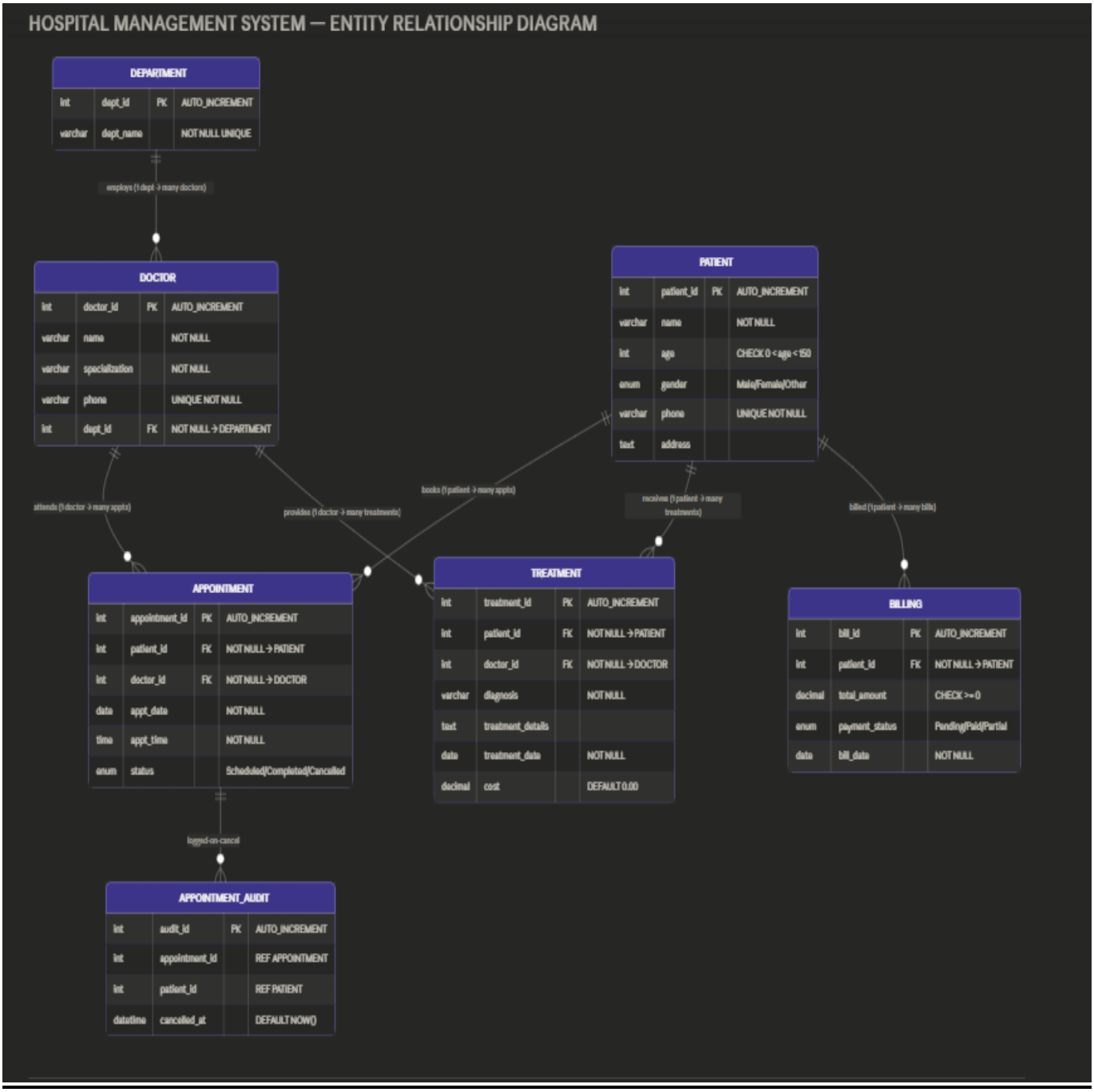
🔷 Relationships:

- Patient ↔ Appointment
- Doctor ↔ Appointment
- Doctor ↔ Department
- Patient ↔ Treatment
- Patient ↔ Billing

🔷 Sample Data:

Sample data was inserted into all tables to:

- Simulate real hospital scenarios
- Test queries and GUI functionality
- Demonstrate system output



Core entities (strong) Associative / weak entities Audit / derived entity

||—o{ one-to-many (mandatory–optional) ||—|| one-to-one (mandatory–mandatory)

PK Primary key | FK Foreign key

CHAPTER 5: DDL COMMANDS & SQL FEATURES

◆ DDL Commands

- CREATE DATABASE
- CREATE TABLE
- Constraints: PRIMARY KEY, FOREIGN KEY, UNIQUE, CHECK

◆ Stored Procedures

- Add patient
- Book appointment (with transaction)
- Generate patient report

◆ Triggers

- Auto-update billing when treatment is added
- Prevent deletion of completed appointments
- Log cancelled appointments

◆ Views

- Patient appointment history
- Pending bills

◆ Transactions

- Ensures atomic operations (appointment + billing)

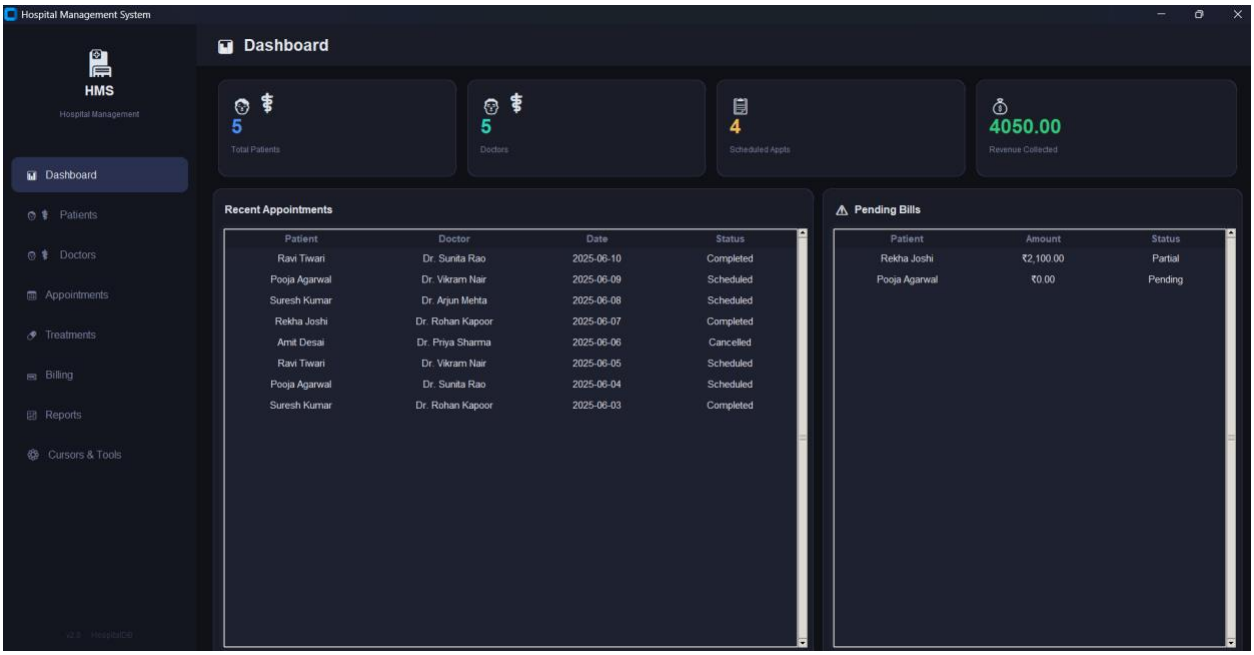
CHAPTER 6: RESULTS & SNAPSHOTS

System Features Demonstrated:

- Patient Management
- Doctor Management
- Appointment Booking
- Treatment Recording
- Billing System
- Reports Dashboard

GUI Features:

- Interactive forms
- Data tables
- Real-time updates
- Report visualization



CHAPTER 7: CONCLUSION, LIMITATIONS & FUTURE WORK

◆ Conclusion

The project successfully demonstrates the implementation of a hospital management system using SQL and GUI. It integrates multiple database concepts into a real-world application.

◆ Limitations

- Limited scalability for very large hospitals
- Basic GUI design
- No authentication system

◆ Future Work

- Add login/authentication system
- Web-based version
- Integration with real hospital systems
- Advanced analytics and dashboards